

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties Of Triangle - Core Application	36 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties Of Triangle - Core Application (Very High Yield — 36 PYQs)

Core Principle: Key problem pattern in Properties Of Triangle.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "Let the area of a $\triangle PQR$ with vertices $P(5,4), Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O(2, \frac{14}{3})$..." [Ans: N/A]
- [2024 Evening Shift] Q#1 (Numerical): "Consider a triangle ABC having the vertices $A(1,2), B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = (\pi/6)$ and $\angle \dots$ " [Ans: 14]
- [2024 Evening Shift] Q#2 (MCQ): "Two vertices of a triangle ABC are $A(3,-1)$ and $B(-2,3)$, and its orthocentre is $P(1,1)$. If the coordinates of the point C are $(\dots)$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 23rd January Morning Shift (Q#1)

Let the area of a $\triangle PQR$ with vertices $P(5,4)$, $Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O(2, (14/5))$ and $C(c,d)$ respectively, then $c+2d$ is equal to

- (A) 3
- (B) $(7/3)$
- (C) 2
- (D) $(8/3)$

Correct Answer: None

Solution Analysis: Equation of line $QR = 5x + 2y + 2 = 0$ Equation of line $PR = 10x - 3y - 38 = 0 \therefore$ Point $R(2, -6)$ Centroid $= ((5 + (-2) + 2)/3, (4 + 4 + (-6))/3) = ((5/3), (2/3))$ $c + 2d = (5/3) + (4/3) = 3$